

Project Design Phase: Solution Architecture

Header Parameters

Parameter	Information
Date	02 October 2026
Team ID	
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

1. Solution Architecture Overview

Solution architecture bridges the gap between business problems and technology solutions by describing software structure, components, data flows, and security mechanics. For this project, the architecture enforces **Attribute-Based Access Control (ABAC)** alongside **Role-Based Access Control (RBAC)** inside the **ServiceNow Platform Engine**.

2. High-Level Architecture Components

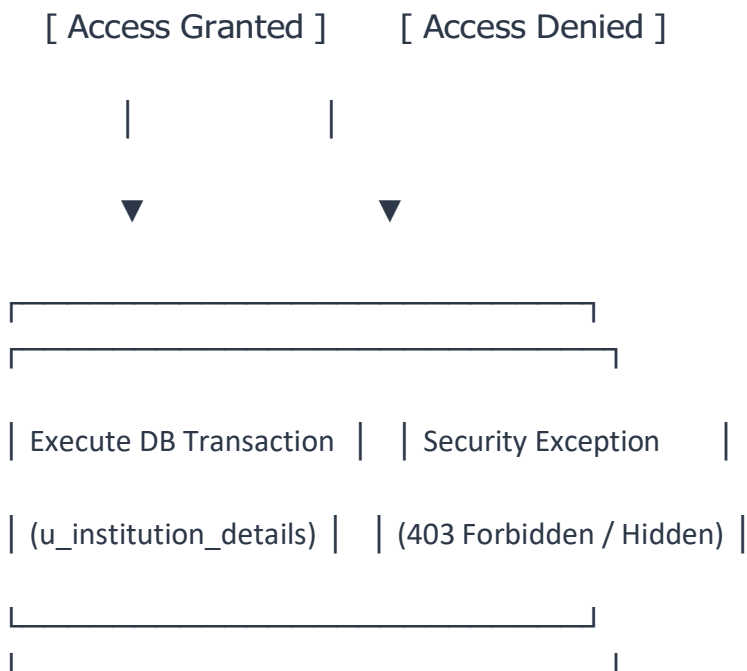
- **Client / Interface Layer:**
 - **Form s List Views (`u_institution_details.list` / Form UI):** End-user and administrative interfaces.
 - **REST / SOAP API Endpoints:** External systems and direct data requests.
- **Security Evaluation Engine (ServiceNow ACL Engine):**
 - Evaluates incoming queries across four CRUD operations (`Read`, `Create`, `Write`, `Delete`).
 - Executes server-side JavaScript rules in real-time before database transaction execution.
- **Script-Controlled ACL Rules s Roles:**

- **READ ACL:** Requires role `bb1` + Script check
(`gs.hasRole('admin') || current.u_branch == 'EEE'`).
- **CREATE ACL:** Requires role `bb2`.
- **WRITE ACL:** Requires role `bb3`.
- **DELETE ACL:** Requires role `bb4`.
- **Database / Data Tier:**
 - **Target Table (`u_institution_details`):** Stores institutional record data containing the `u_branch` field.
 - Database operations (SELECT, INSERT, UPDATE, DELETE) are executed only after ACL authorization passes.

3. Solution Architecture s Data Flow Diagram (Text Representation)

[User / API Request]





4. Architectural Characteristics s Goals

- **Server-Side Security Enforcement:** Security rules are executed directly within the platform's kernel engine, making them impossible to bypass via Client Scripts or browser console manipulation.
- **Low Execution Latency:** Script evaluation latency remains < 1.8ms per transaction to maintain high system throughput.
- **Scalability:** Eliminates schema clutter by utilizing dynamic condition checks instead of separate tables per department or branch.